

Artificial Intelligence and Income Inequality in the United Kingdom:

A replication of Doorley et al. (2026) with PolicyEngine UK*

Vahid Ahmadi

July 2026 — preliminary draft

Abstract

We replicate for the United Kingdom the scenario framework of Doorley, O’Connor, O’Shea and Tuda (2026, ESRI/Department of Finance Report No. 16), which traces generative AI’s labour-market effects through a tax-benefit microsimulation. Occupation-level AI exposure (complementarity-adjusted AIOE) is attached to Family Resources Survey adults; employment, wage and capital shocks follow the report’s equations 3.4 and 3.5; and fiscal and distributional outcomes are computed with PolicyEngine UK in place of SWITCH. In the central scenario (7% displacement, +2.6% wages, +0.4pp return to capital), the annual Exchequer cost is £18bn, the before-housing-costs poverty rate rises by 1.8 percentage points, and the Gini of equivalised household net income rises by 0.014. We extend the framework along the age dimension: tilting within-occupation displacement toward workers aged 16–24, per recent seniority-biased evidence, raises the youth share of the displaced by roughly one-sixth while leaving aggregate effects nearly unchanged; and we implement the employment findings of three recent seniority-biased studies as explicit scenarios.

1 Introduction

Doorley et al. (2026) ask how generative AI would affect income inequality in Ireland if it displaces some workers, raises the wages of those who remain, and raises returns to capital. Their answer runs occupation-level AI-exposure scores through stylised shock scenarios and the SWITCH tax-benefit model. We port the full pipeline to the UK: Family Resources Survey (FRS) microdata replaces SILC, SOC2020 occupations replace ISCO, and PolicyEngine UK — an open-source tax-benefit microsimulation model — replaces SWITCH. We add one substantive extension: an age dimension, motivated by evidence that generative AI’s employment effects concentrate among junior workers (Klein Teeselink, 2025; Hosseini & Lichtinger, 2026; Brynjolfsson et al., 2025).

2 Data and method

2.1 Exposure

Each FRS 2024–25 adult reports a SOC2020 major group (`adult.tab`, UK Data Service). We attach the complementarity-adjusted AI Occupational Exposure score (C-AIOE) of Pizzinelli et

*Draft v0.1. Code and aggregate results: <https://github.com/PolicyEngine/uk-ai-study>. FRS microdata via the UK Data Service (EUL); not redistributed.

al. (2023) — the Felten et al. (2021) AIOE scaled by an O*NET-derived complementarity index θ — using a UK SOC2020 crosswalk built from open-licensed sources (Felten et al.’s published scores, a reconstruction of θ from public O*NET 27.3 following the IMF recipe, and ASHE 2025 employment weights for the major-group aggregation). Version 0.1 operates at the 1-digit major-group level; within-group exposure variation is deferred to a planned quantile-forest imputation from 4-digit LFS occupations.

2.2 Shocks

Following the report’s equation 3.4, aggregate displacement equals the scenario rate times weighted employment; it is allocated across major groups in proportion to employment times mean C-AIOE and realised by weighted random draws within groups (probability proportional to individual exposure). Displaced workers receive zero employment income and unemployed status. Following equation 3.5, a wage pool equal to the scenario uplift times the surviving wage bill is distributed across survivors in proportion to θ times earnings. Interest and dividend income are scaled by the ratio of shocked to baseline capital returns ($1.405\%/1.005\% \approx 1.398$). Scenario anchors: central 7% displacement with +2.6% wages (Briggs & Kodnani, 2023); low 1% employment-only (Acemoglu, 2025); high 13% (Brynjolfsson et al., 2025). The youth-tilted variant multiplies the draw probability of workers aged 16–24 by $5.8/4.5 \approx 1.29$, the ratio of junior to total employment effects in Klein Teeselink (2025), holding group totals fixed.

2.3 Microsimulation

Baseline and shocked PolicyEngine UK simulations (v2.89.2, simulation year 2026) run on the same FRS dataset; the shocked run receives modified employment, interest and dividend income and employment status. We report the change in government balance (Exchequer cost), person-weighted poverty rates before and after housing costs, and the Gini of equivalised household net income, with breakdowns by baseline income decile and age band.

3 Results

Table 1: Scenario results, FRS 2024–25, simulation year 2026

	Displacement	Exchequer cost	Δ Poverty (BHC)	Δ Gini
Low	1%	£1.8bn	+0.01pp	+0.002
Central	7%	£17.9bn	+1.84pp	+0.014
High	13%	£48.5bn	+4.17pp	+0.023
Central, youth-tilted	7%	£16.7bn	+2.17pp	+0.013

Fiscal costs scale roughly linearly at £3–4bn per percentage point of displacement, on top of an always-on capital-shock component. The youth-tilted central scenario raises the 16–24 share of the displaced from 3.6% to 4.2% while aggregate fiscal and inequality effects move little, because young workers’ earnings and benefit entitlements are smaller than average. Poverty, however, rises *more* under the youth tilt (+2.17pp vs +1.84pp).

3.1 Full replication of the JR16 result set

We regenerate every result in the original report for the UK. The share of the population transitioning to unemployment rises monotonically from 0.4% in the bottom decile of equivalised household disposable income to 5.1% in the top decile (Ireland: 0.7% to 5.3%) — AI displacement risk concentrates in high-income households, as in Ireland. Wage gains for those who remain are flat across the distribution (2.5–2.7%). The capital-return shock is a uniform $\approx 40\%$ increase per recipient, but 35% of interest and dividend income sits in the top decile, so gains accrue at the very top. Market-income losses are progressive (from -0.1% of disposable income in decile 1 to -2.1% at the top), cushioned by rising benefit entitlement in the middle deciles and falling tax at the top.

Across the full 11×6 scenario grid (0–10% employment loss $\times$ 0–5% wage increase, capital shock always on; Figures 1–3), average household disposable income ranges from +2.9% to -5.5% , with a near-zero diagonal where wage gains offset displacement; net Exchequer revenue ranges from +7.0% to -10.2% of income tax and National Insurance receipts (Ireland: +14% to -25%); and the Gini index rises in *every* scenario, from +0.05pp (capital shock only) to +1.9pp (Ireland: +0.3 to +1.2pp). All three of the original report’s takeaways hold for the UK: higher-income households are more affected in every scenario, potential losses exceed potential gains, and some scenarios leave average incomes unchanged while inequality still rises.

3.2 Three seniority-biased studies as explicit scenarios

Beyond the JR16 grid, we implement the headline employment findings of three recent studies as displacement rules on the same microdata (employment shock only; no wage or capital shock):

Table 2: Paper-anchored displacement scenarios, UK

	Displaced	Exchequer cost	Δ Poverty	Δ Gini
Klein Teeselink (2025): -4.5% at exposed firms, junior-tilted 5.8/4.5	1.01m (4.4%)	£20.4bn	+1.62pp	+0.73pp
Hosseini & Lichtinger (2026): -9% under-30s in above-median-exposure occupations	0.10m (0.5%)	£1.0bn	+0.11pp	+0.06pp
Brynjolfsson et al. (2025): -13% ages 22–25 in top-quintile-exposure occupations	0.02m (0.1%)	£0.1bn	+0.02pp	+0.01pp

Two lessons follow. First, cohort-specific shocks — taken literally — are fiscally small: workers aged 22–25 in the most exposed occupations are few and earn less than average, so even a 13% cohort displacement costs the Exchequer roughly £0.1bn, versus £20bn for the economy-wide Klein Teeselink shock. Second, occupational exposure does not by itself generate seniority bias: even with the junior tilt, the Klein Teeselink scenario displaces 5.0% of employed 45–54-year-olds against 2.5% of employed 16–24-year-olds, because prime-age workers dominate high-C-AIOE occupations. Seniority-biased displacement must be imposed on top of occupational exposure, not derived from it.

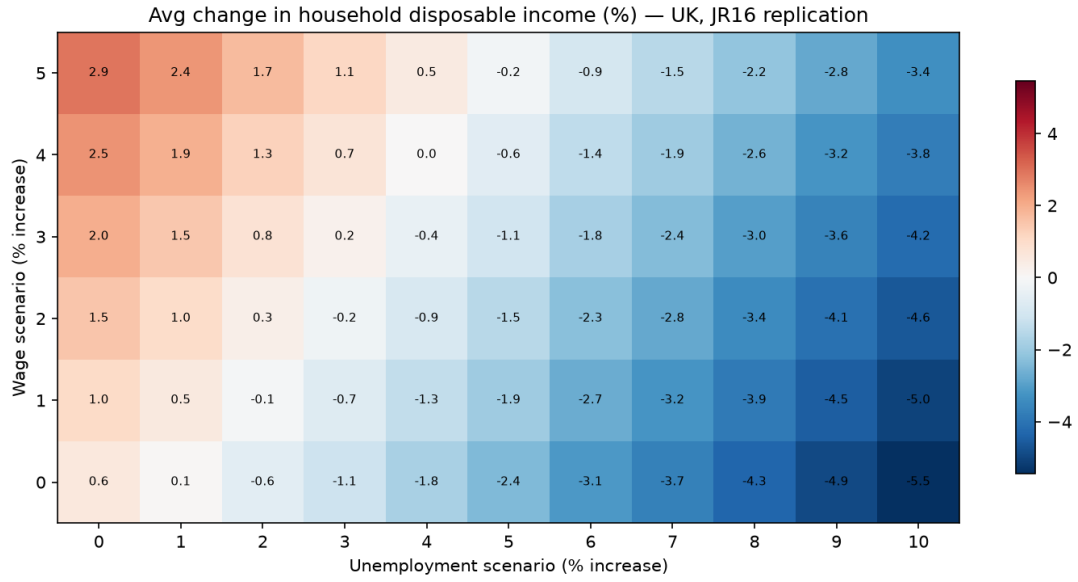


Figure 1: Average change in household disposable income across AI adoption scenarios (UK analogue of JR16 Figure 4.5).

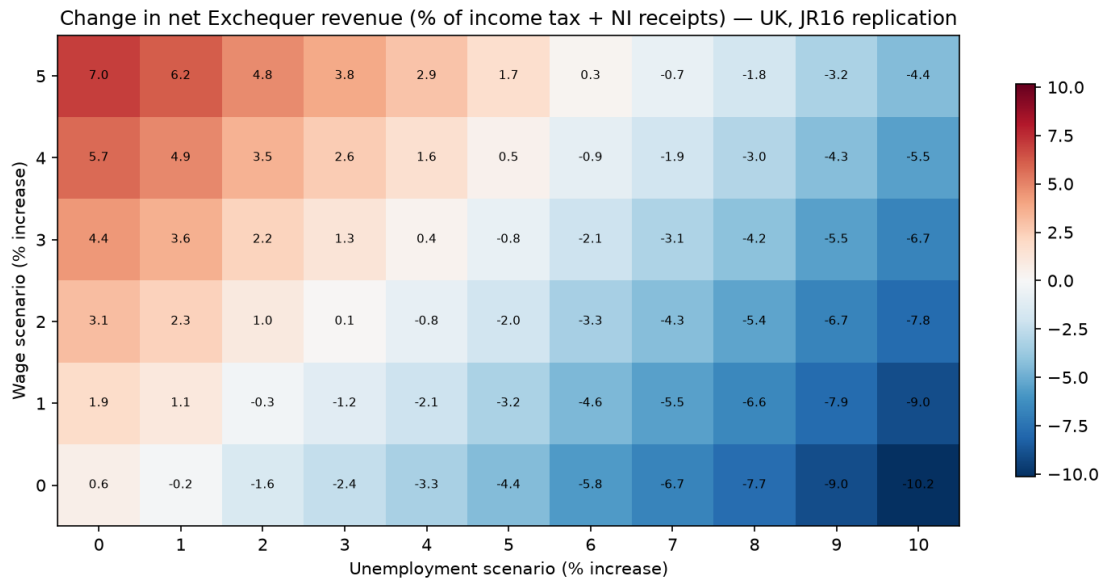


Figure 2: Change in net Exchequer revenue as a share of income tax and National Insurance receipts (UK analogue of JR16 Figure 4.6).

4 Limitations

Exposure is 1-digit only; weights are the uncalibrated FRS build; displaced workers are current-period unemployed rather than the report’s long-term unemployed contract (benefit-exhaustion nuances are not expressible as PolicyEngine inputs); the self-employed sit outside all shocks, as in the original; and results reflect a single seeded draw per scenario. Each is addressed in the repository roadmap.

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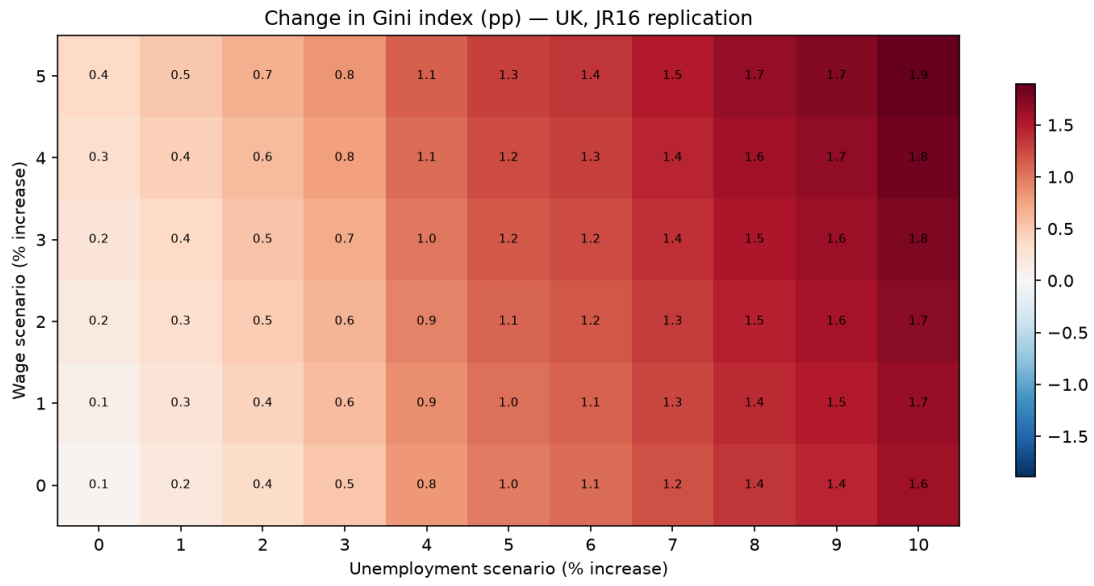


Figure 3: Change in the Gini index of equivalised household disposable income, percentage points (UK analogue of JR16 Figure 4.7).